1. (a) Prove Fermat's Little Theorem: if p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \mod p$.
   (b) If G is a group of prime order p, then G is cyclic.
   (c) A nontrivial group G has no nontrivial proper subgroups if and only if G is finite and of order p where p is prime.

(a) Claim ①: $\forall a \in \mathbb{Z}^+$ if p is prime, then $a^p \equiv a \mod p$.

Proof: Take arbitrary prime p.

We prove it by induction on a.

Basic step: $1^p = 1$, So $1^p \equiv 1 \mod p$.

Inductive step: Assume $a^p \equiv a \mod p$, we can show then $(a+1)^p \equiv (a+1) \mod p$

$$(a+1)^p = \sum_{k=0}^{p} \binom{p}{k} a^k = \sum_{k=0}^{p} \frac{p!}{k!(p-k)!} a^k$$

$$= \left(\sum_{k=1}^{p-1} \frac{p!}{k!(p-k)!} a^k\right) + 1 + a^p$$

For each $1 \le k \le p-1$, $p-k \le p-1$ and $p-k \ge 1$,

so $p \nmid$ any one of the terms in $k!(p-k)!$ ①

Since p is prime, $p \nmid k!(p-k)!$

(otherwise p must divide one of the terms in $k!(p-k)!$ which contradicts with ①)

Therefore $\forall\ 1 \leq k \leq p-1$, $\dfrac{(p-1)!}{k!(p-k)!}$ is still integer

$$\text{So } (a+1)^{p-1} = p\left(\sum_{k=1}^{p-1} \frac{(p-1)!}{k!(p-k)!} a^k\right) + 1 + a^p$$

$$\text{where } p \,\Big|\, \left(\sum_{k=1}^{p-1} \frac{(p-1)!}{k!(p-k)!} a^k\right)$$

$$\text{i.e. } p \,\Big|\, \left((a+1)^p - (a^p + 1)\right)$$

Therefore $(a+1)^p \equiv a^p + 1 \pmod p$

Since $a^{p-1} \equiv 1 \pmod p \Rightarrow \underline{a^p \equiv a \pmod p}$

$\underline{\text{So } (a+1)^p \equiv a+1 \pmod p}$

Then we have pro d proved the Claim ①

<u>Claim ② : Following claim ①, if $p \nmid a$, then $a^{p-1} \equiv 1 \pmod p$</u>

<u>Proof</u> Let p be arbitrary prime and take arbitrary $a \in \mathbb{Z}^+$ with $p \nmid a$

By claim ①, $a^p \equiv a \pmod p$

So $p \mid a^p - a \Rightarrow p \mid a(a^{p-1} - 1)$

Since p is prime $\Rightarrow p \mid a$ or $p \mid a^{p-1} - 1$

Since $p \nmid a \Rightarrow p \mid a^{p-1} - 1$.

So $a^{p-1} \equiv 1 \pmod p$

Thereby we have proved the Claim ② and combining claim ① and ② we have proved the Fermat's Little Thm.

(b) Proof. Assume $|G|$ is prime, so $|G| \geq 2$, so therefore must exists non-identity elements in G (since there can only be exactly one identity).

So Select arbitrary non-identity element $a \in G$

Then $|\langle a \rangle| \geq 2$ Since at least $a, a^2 \in \langle a \rangle$

($a \neq e$, otherwise $a^{-1}a^2 = a^{-1}a \Rightarrow a = e$)

By the Lagrange Thm, $|G| = |\langle a \rangle| [G : \langle a \rangle]$

So $|\langle a \rangle| \mid |G|$

Since $|G|$ is prime $\Rightarrow |\langle a \rangle| = 1$ or G

Then since $|\langle a \rangle| \geq 2$, $|\langle a \rangle| = |G|$ which means $\langle a \rangle = G$

(c) First we prove on the backward direction:

Assume $|G|$ is finite and is prime.

Then $\forall$ subgroup $K \subseteq G$, by Lagrange's Thm $|K| \mid |G|$.

Since $|G|$ is prime $\Rightarrow |K| = 1$ or $|G|$

So K is either trivial or is G itself

Therefore G has no nontrivial subgroup.

Then we prove on the forward direction.

We do it by contradiction: Assume G has no nontrivial proper subgroup, but G is of composite finite order (Case ①) or G has infinite order. (Case ②)

Case ①: $|G| = mn$ for some prime m and $n \geq 2$

We know by book Thm 8.6: $\forall x \in G$, $x^{|G|} = e$.

So we pick arbitrary $x \in G$ s.t. $x \neq e$

Then $x^{mn} = e$

If $x^m = e \Rightarrow$ then $\langle x \rangle$ whose order $\leq m < |G|$ is a nontrivial proper subgroup of $G \Rightarrow$ contradicts.

If $x^m \neq e \Rightarrow \langle x^m \rangle$ whose order $\leq n < |G|$ is a nontrivial proper subgroup of $G \Rightarrow$ contradicts.

Case ②: G has infinite order.

We select arbitrary non-identity $g \in G$

And consider subgroup $\langle g \rangle, \langle g^2 \rangle \leq G$

① If $g \in \langle g^2 \rangle \Rightarrow g = (g^2)^n = g^{2n}$ for some integeter n

So $g^{2n-1} = e \Rightarrow |\langle g \rangle| \leq 2n-1$

Then $\langle g \rangle$ is a non-trivial proper subgroup of G
$\Rightarrow$ contradicts.

② If $g \notin \langle g^2 \rangle$, Then it is sure that $\langle g \rangle \neq \langle g^2 \rangle$

if $g^2 = e$ then $\langle g \rangle = \{e, g\}$ is ensured a non-trivial proper subgroup of G

Otherwise $\langle g^2 \rangle \subsetneq G$ is ensured a non-trivial proper subgroup of G
$\Rightarrow$ contradicts.

By Case ①②, we have proved on the forward direction and finished the proof of: a non-trivial group G has no non-trivial proper subgroup iff G is infinite or $|G| = p$ which is prime.

2. For each of the following parts, K is a subgroup of the group G. Write down every element of every (distinct) right coset AND every distinct left coset. You do not need to prove that you have found every coset.

(a) $K = \{r_{0^\circ}, r_{90^\circ}, r_{180^\circ}, r_{270^\circ}\}$, G is D_4, the set of symmetries of the square.
For the sake of notation, denote the reflections of the square as s_v, s_h, s_{NW} and s_{NE}.

(b) $K = \{e, (12)\}$, $G = S_3$

(c) $K = \left\langle \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\rangle$, $G =_2 (\mathbb{Z}_2)$. For your convenience, the elements of $\overset{GL}{}_2(\mathbb{Z}_2)$ are given below.
You may use the given letters to refer to them:

$$GL_2(\mathbb{Z}_2) = \left\{ I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, a = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, b = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, c = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, d = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} f = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \right\}$$

(d) $K = \langle 5 \rangle$, $G = \mathbb{Z}_{12}^{\times}$

Sol (a) There are two left/right cosets:

Left cosets:

1. $(r_0 K = r_{90} K = r_{180} K = r_{270} K =) K$

2. $s_v K = \{s_v, s_h, s_{NW}, s_{NE}\}$
$(= s_v K = s_h K = s_{NW} K = s_{NE} K)$

Right cosets:

1. $(K r_0 = K r_{90} = K r_{180} = K r_{270} =) K$

2. $K s_v (= K s_h = K_{NW} = K_{NE}) = \{s_v, s_h, s_{NW}, s_{NE}\}$

(b) $G = S_3 = \{e, (12), (13), (23), (123), (132)\}$

There are three left/right cosets.

Left cosets:

1. $(13)K = \{(13), (123)\}$
2. $(23)K = \{(23), (132)\}$
3. K itself.

Right cosets:

1. $K(13) = \{(13), (132)\}$
2. $K(23) = \{(23), (123)\}$
3. K itself

(c) $K = \{a, I\}$

There are three left/right cosets.

Left cosets:

1. $aK = IK = K = \{a, I\}$
2. $fK = \{\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, f\} = \{d, f\} = dK$
3. $bK = \{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, b\} = \{c, b\} = cK$

Right cosets:

1. $KI = Ka = K = \{a, I\}$
2. $Kf = \{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, f\} = \{c, f\} = Kc$
3. $Kd = \{\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, d\} = \{b, d\} = Kb$

(d) $G = \{1, 5, 7, 11\}$

$K = \langle 5 \rangle = \{5, 1\}$

There are two left / right cosets.

Left cosets:

1. $K = \{5, 1\}$
2. $7K = \{11, 7\}$

Right cosets:

1. $K = \{5, 1\}$
2. $K7 = \{11, 7\}$

3. Any group G acts on itself by conjugation: $g \cdot h = ghg^{-1}$. The orbits of this action are called **conjugacy classes**.
   1. Show $h \in Z(G)$ if and only h is a fixed point of the conjugation action.
   2. Show a subgroup H of G is normal if and only if it is a disjoint union of conjugacy classes.
   3. Describe the partition of $\mathcal{S}_5$ into its conjugacy classes.
   4. Show that the only nontrivial normal subgroup of $\mathcal{S}_5$ is $\mathcal{A}_5$.[1]

[1] Hint: By (b), a normal subgroup is a union of conjugacy classes, one of which is the identity. Use the sizes of these conjugacy classes from (c), plus Lagrange's Theorem, to narrow down the list, and finally show that on your shortlist, the only collection closed under products is $\mathcal{A}_5$.

1. Pf The forward direction: Assume h is a fixed point of the conjugation action,

Then $\forall g \in G$, $ghg^{-1} = h$

$\overset{\cdot g^{-1} \text{ on both sides}}{\Longrightarrow} gh = hg$

So $gh = hg$ for all $g \in G$ $\Rightarrow h \in Z(G)$

The backward direction: Assume $h \in Z(G)$,

Then $\forall g \in G$, $gh = hg$

So take arbitrary $g \in G \Rightarrow gh = hg$

$\Rightarrow ghg^{-1} = hgg^{-1}$

$\Rightarrow ghg^{-1} = h$

Therefore h is a fixed point of the conjugation action.

Thereby we have proved: h is a fixed point of the conjugation action iff $h \in Z(G)$

2. Pf. Forward direction: Assume H is a disjoint union of conjugacy classes. i.e.

$H = \{g h_1 g^{-1} \mid g \in G\} \sqcup \{g h_2 g^{-1} \mid h_2\} \ldots \sqcup \{g h_n g^{-1} \mid h_n\}$

for some $h_1, h_2, \ldots, h_n \in G$.

Select arbitrary $g \in G$ and fix it.

Then take arbitrary $x \in H \Rightarrow x \in O(h_i)$ for some $h_i \in \{h_1, \dots, h_n\}$,

So $\underbrace{gxg^{-1} = g \cdot x \in O(h_i)} \subseteq H$

So $g^{-1}xg \in H$

Therefore $\underbrace{\forall g \in G,\ gHg^{-1} \subseteq H}$

So by Thm 8.11, H is a normal subgroup of G.

Backward direction: Assume H is a normal subgroup of G.

Then take arbitrary $g \in G$

By Thm 8.11, $gHg^{-1} \subseteq H$, so

$\forall h \in H$, $\underbrace{ghg^{-1} \in H} \Rightarrow g \cdot h \in H$

Since g is arbitrary $\Rightarrow \underbrace{O(h) \subseteq H}$

Therefore we can conclude that for all $h \in H$, $O(h) \subseteq H$. So H is a union of conjugate classess. Since we have proved that orbits are either disjoint or identity, H is a disjoint union of conjugate classess

Thereby we have proved that a subgroup $H \subseteq G$ is normal iff it is a disjoint union of conjugate classes

3. $O(e) = \{e\}$, order $= 1$

$O((12)) = \{$All 2-cycles$\}$, order $= \binom{5}{2} = \underline{10}$

$O((123)) = \{$All 3-cycles$\}$, order $= \binom{5}{3} \times 2 = \underline{20}$

$O((1234)) = \{$All 4-cycles$\}$, order $= \binom{5}{4} \times 3! = \underline{30}$

$O((12345)) = \{$All 5-cycles$\}$, order $= 4! = \underline{24}$

$O((12)(34)) = \{$All two disjoint transitions$\}$,

order $= \frac{1}{2} \times \binom{5}{2} \times \binom{3}{2} = \underline{15}$

$O((12)(345)) = \{$All 2+3-disjoint cycles$\}$,
(3+2, the same)

order $= \binom{5}{3} \times 2 = \underline{20}$

(These union to be S_5, with orders sum up to 120)

4. Let K be a nontrivial subgroup of S_n

First, $\underline{e \in K}$ by the definition of subgroup.

By 2, K is a disjoint union of conjugacy classes

so $\underline{\{e\}}$ is one of conjugacy classes that form K.

By Lagrange's Thm, $|K| \,|\, |S_n|$, so $\underline{|K| \,|\, 120}$

$\underline{\text{Since } K \text{ is non-trivial, } K \neq \{e\}.}$

So there must be more conjugacy classes in the disjoin unions

Since $|\{e\}|=1$ and $|K| \mid 120$

{all 5-cycles} whose order is 24 must be one of the conjugacy classes since otherwise $|K|$ cannot divide 120.

Now $|K| \geq 25$

Note that then $|K|$ can only be 30, 40 or 60 to divide 120

So {all two disjoint transitions} whose order is 15 must be one of the conjugacy classes

Now there are three possibilities:

① $K = \{e\} \sqcup \{\text{all 5-cycles}\} \sqcup \{\text{all 2-disjoint transitions}\}$

② $K = \{e\} \sqcup \{\text{all 5-cycles}\} \sqcup \{\text{all 2-disjoint transitions}\} \sqcup \{\text{all 3-cycles}\}$

③ $K = \{e\} \sqcup \{\text{all 5-cycles}\} \sqcup \{\text{all 2-disjoint transitions}\} \sqcup \{\text{all 2+3-disjoint cycles}\}$

Note that another necessary condition for the normal subgroup is that K is a subgroup (closed under operation and inverse)

For ①, since $(34)(12)(12345) = (245) \notin K$,

it is not a subgroup. $\Rightarrow$ impossible.

For ③, since $(34)(123)(12345) = (1452) \notin K$,

it is not a subgroup. $\Rightarrow$ impossible.

Therefore only ② can be a subgroup.

Note that K in ② is A_5, since $\forall a \in S_5 \setminus K$

is odd and $\forall b \in K$ is even. A_5 is a subgroup of S_5.

And by (2), K is the only non-trivial normal

subgroup of S_5.

4. Let p be a prime, and G be a finite group with $p \mid |G|$. Consider the set

$$X = \{(g_1, \dots, g_p) \in \underbrace{G \times \cdots \times G}_{p\text{-times}} \mid g_1 g_2 \cdots g_p = e\}.$$

The group $\mathbb{Z}_p$ acts on X by rotating elements: $[i]_p \cdot (g_1, \dots, g_p) = (g_{1+i}, \dots, g_p, g_1, \dots, g_i)$.

1. Show that X has $|G|^{p-1}$ elements, so $p \mid |X|$.
2. Show that the orbits of the action of $\mathbb{Z}_p$ on X either have 1 or p elements, and the orbits of order 1 are either $(e, e, \dots, e)$ or of the form $(g, g, \dots, g)$ with $|g| = p$.
3. Show that G contains an element of order p.

1. Pf. For any choice of $(g_1, g_2, \dots, g_{p-1})$, by the existence and uniqueness of inverse of an element in group,

$\exists! g_p$ s.t. $g_1 g_2 \cdots g_p = e$, which is fixed

Therefore since we have $|G|^{p-1}$ choices of $(g_1, g_2, \dots, g_{p-1})$, we have $|X| = |G|^{p-1}$.

Since $p \mid |G|$ and $|X| = |G| \cdot |G|^{p-2}$, $p \mid |X|$.

2. Pf. $\forall x \in X$

By the Orbit-Stablizer Theorem,

$|\text{Orbit}(x)| \cdot |\text{Stab}(x)| = |\mathbb{Z}_p| = p$

① Consider if $x = (g, g, g, \dots, g)$ for some $g \in G$

Then $\forall y \in \mathbb{Z}_p$, $y \cdot x = x \Rightarrow \text{Stab}(x) = \mathbb{Z}_p$

$\Rightarrow |\text{Stab}(x)| = p \Rightarrow |\text{Orbit}(x)| = 1$

Note that in this case, either $g=e$ or $|g|=p$

This is because $(g,g,..g) \in X$ which means that $g^p=e$.

So either $g=e$, or $|g|=p$.

(* $|g|$ can not be $<p$, since if so then $g^p = g^{|g|\cdot a}=e$ for some $a \in \mathbb{Z}$, contradicts with p is prime)

② Otherwise, $x=(g_1,g_2,\dots,g_p)$ where at least some $g_i, g_j \in \{g_1,g_2,\dots,g_p\}$ are different

Therefore only when $y=[0]_p$, $y \cdot x = x$

$\Rightarrow Stab(x) = \{[0]_p\} \Rightarrow |Stab(x)|=1$

$\Rightarrow |Orbit(x)|=p$

Thereby we have proved: the orbits of the action of $\mathbb{Z}_p$ on X either have 1 or p elements, and the orbits of order 1 are either $(e,e,\dots,e)$ or the form of $(g,g,g,\dots,g)$ where $|g|=p$.

(3) Since ① every element of X belongs to exactly one orbit

② we have proved that $p \mid |X|$, so $|X|=pm$ for some $m \in \mathbb{Z}$

③ we know $\forall x \in X$, $|O(x)|=1$ or p

By reducing elements of X such that $|O(x)| = p$ from X, we get $X' = \{(g,g,\dots,g) \mid |g| = p\} \cup \{(e,e,\dots,e)\}$

with $|X'| = mp - np$ for some $n \in \mathbb{Z}$

so $p \mid |X'|$

Since there e must exist, $(e,e,\dots,e) \in X$

So there must be at least $p-1$ $(p \geq 2)$ element of the form $(g,g,g,\dots,g) \in X$ such that $|g| = p$

Therefore there there must exist $g \in G$ such that $|g| = p$.